

EE3621: Digital Signal Processing

Lecture 4: Frequency Response, Magnitude/Phase & Group Delay of LTI Systems (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 08:00 (8 mins):** LTI Eigenfunctions: Why complex exponentials are unique. Derivation of $H(e^{j\omega})$.
 - **08:00 – 15:00 (7 mins):** Magnitude & Phase responses, Rectangular vs. Polar forms.
 - **15:00 – 25:00 (10 mins):** Phase Delay vs. Group Delay. Narrowband input derivation (Carrier vs. Envelope delay).
 - **25:00 – 33:00 (8 mins):** Signal Impacts in Time/Frequency Domains & Voltage Signal Characteristics.
 - **33:00 – 38:00 (5 mins):** First-order IIR Filter Example: Detailed algebraic derivation of Group Delay.
 - **38:00 – 40:00 (2 mins):** Checkpoints & Concept Recap.
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2 Eigenfunctions & The Frequency Response

Complex exponentials $e^{j\omega n}$ are eigenfunctions of linear time-invariant systems. Let the input be $x[n] = e^{j\omega n}$ for $-\infty < n < \infty$. The output $y[n]$ is:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = e^{j\omega n} \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \quad (1)$$

We define the complex scaling factor (eigenvalue) as the frequency response $H(e^{j\omega})$:

$$y[n] = H(e^{j\omega})e^{j\omega n} \quad \text{where} \quad H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \quad (2)$$

3 Magnitude & Phase Responses

The frequency response can be written in polar form:

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)} \quad (3)$$

where $|H(e^{j\omega})|$ is the magnitude response and $\theta(\omega) = \angle H(e^{j\omega})$ is the phase response (Figure 1). For a sinusoidal input $x[n] = A \cos(\omega_0 n + \phi)$, the output is:

$$y[n] = A|H(e^{j\omega_0})| \cos(\omega_0 n + \phi + \theta(\omega_0)) \quad (4)$$

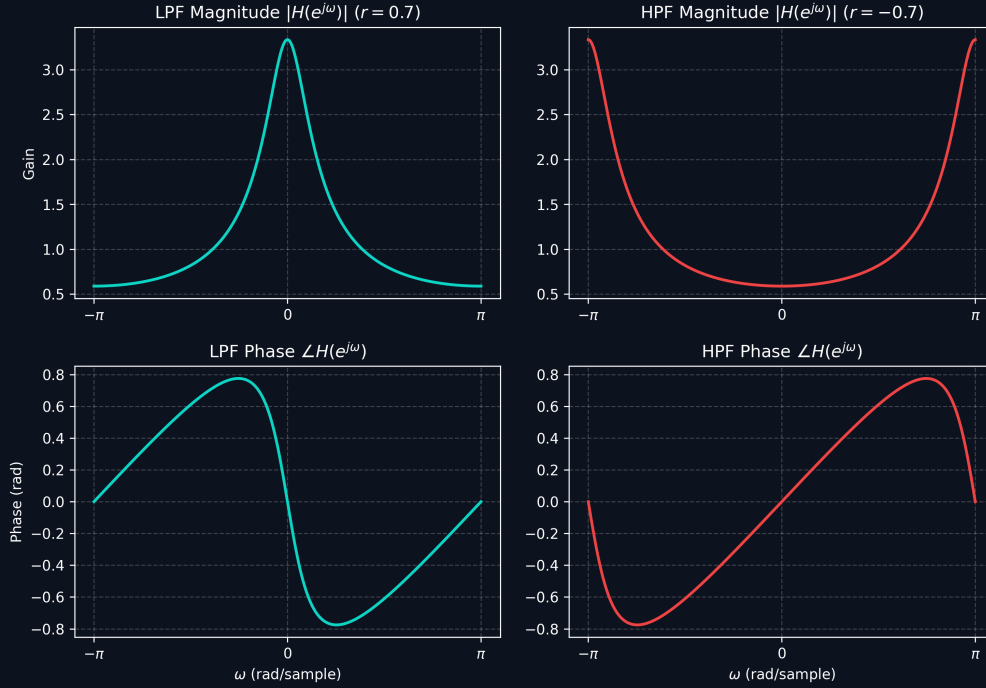


Figure 1: Magnitude and phase responses for first-order digital low-pass and high-pass filters.

4 Phase Delay vs. Group Delay

4.1 Mathematical Definitions & Intuitive Meaning

Concept	Simplest Intuitive Meaning	Formula	Everyday Analogy
Phase Delay (τ_p)	Time delay of a SINGLE pure sine wave	$\tau_p(\omega) = -\frac{\theta(\omega)}{\omega}$	Delay of a runner's stride rhythm
Group Delay (τ_g)	Time delay of the OVERALL ENVELOPE / PACKET	$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega}$	Delay of the entire group / pack of runners

Table 1: Intuitive distinction between Phase Delay and Group Delay.

4.1.1 Intuition Behind Phase Delay (τ_p)

- **Core Question:** *"How long does one individual frequency component take to travel through the filter?"*
- **Derivation:** Sinusoidal phase shift is $\theta(\omega) = -\omega \cdot \tau_p \implies \tau_p(\omega) = -\frac{\theta(\omega)}{\omega}$ (in samples or seconds).

4.1.2 Intuition Behind Group Delay (τ_g)

- **Core Question:** *"How long does the overall shape, modulation burst, or pulse envelope take to pass through the filter?"*
- **Why Derivative $\frac{d\theta}{d\omega}$?** Two nearby frequencies (ω_1, ω_2) interact to form a beat envelope. The delay of this envelope is $\text{Delay} \approx -\frac{\Delta\theta}{\Delta\omega}$. Taking $\Delta\omega \rightarrow 0$ yields the exact derivative

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega}.$$

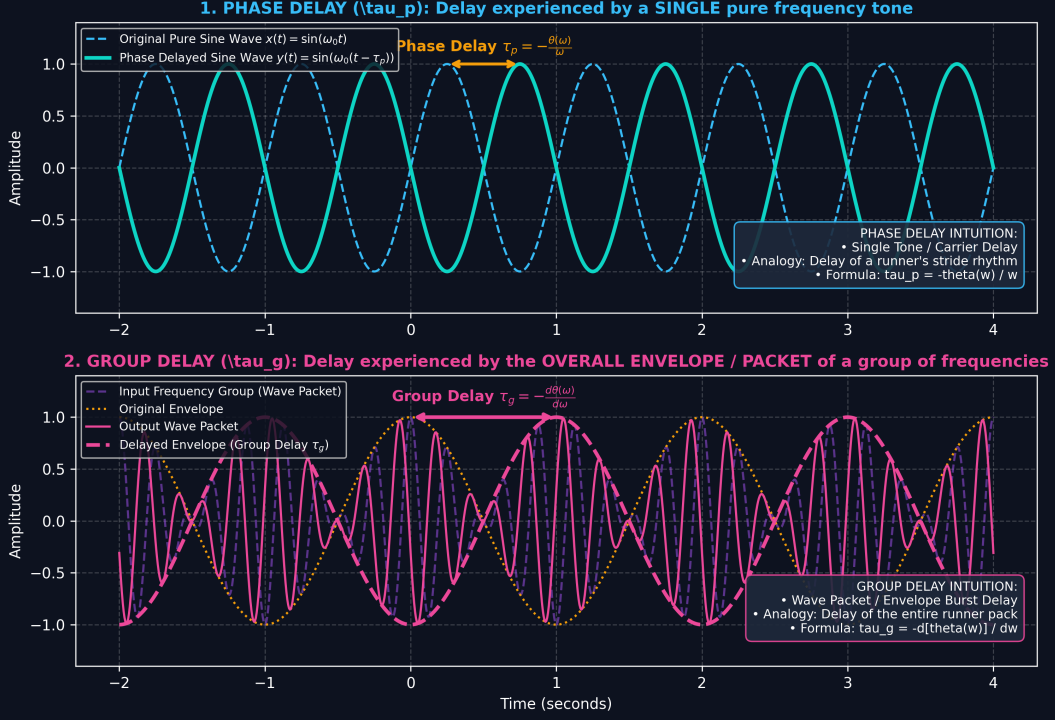


Figure 2: Demonstrative illustration: Phase Delay (τ_p) vs. Group Delay (τ_g) side-by-side comparison with intuitive callout boxes and formulas.

4.2 Narrowband Envelope Propagation

Consider a narrowband wave packet input:

$$x[n] = s[n] \cos(\omega_0 n) \quad (5)$$

Expanding the phase response $\theta(\omega)$ about ω_0 :

$$\theta(\omega) \approx \theta(\omega_0) - \tau_g(\omega_0)(\omega - \omega_0) = -\omega_0 \tau_p(\omega_0) - \tau_g(\omega_0)(\omega - \omega_0) \quad (6)$$

The resulting system output is:

$$y[n] \approx |H(e^{j\omega_0})| s[n - \tau_g(\omega_0)] \cos(\omega_0(n - \tau_p(\omega_0))) \quad (7)$$

Thus, the modulation envelope $s[n]$ is delayed by the group delay $\tau_g(\omega_0)$, while the carrier fine structure is delayed by the phase delay $\tau_p(\omega_0)$ (Figure 2).

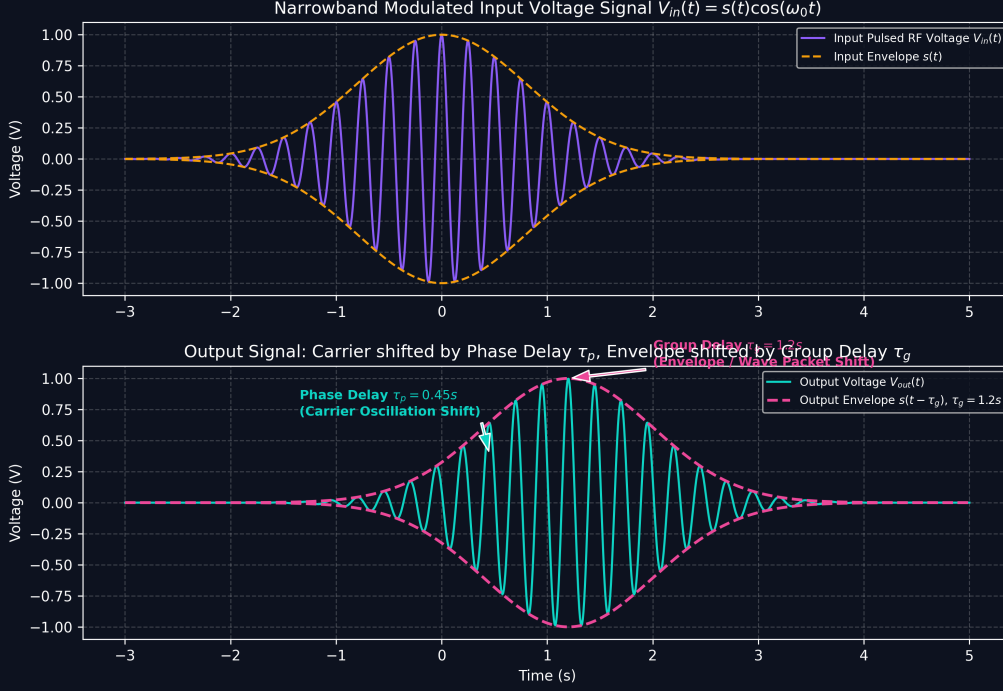


Figure 3: Carrier phase shift vs. Modulation envelope shift under phase delay and group delay.

5 Impact of Phase & Group Delay in Time and Frequency Domains

- **Single Pure Sinusoid (ω_0):** In the time domain, the waveform experiences a uniform time shift of $\tau_p(\omega_0)$ samples without shape distortion. In the frequency domain, the spectral line acquires a phase shift $\theta(\omega_0) = -\omega_0\tau_p(\omega_0)$.
- **Narrowband Signal (Modulated Wave Packet):** In the time domain, the carrier shifts by phase delay τ_p , while the envelope packet shifts by group delay τ_g . In the frequency domain, the local phase spectrum around ω_0 is linear with slope $-\tau_g(\omega_0)$.
- **Broadband Multi-Harmonic Signal (Linear Phase):** When $\tau_g(\omega) = \alpha$ (constant across all frequencies), transmission is **distortionless**. The entire composite signal is shifted by α samples in time with zero shape change.
- **Broadband Multi-Harmonic Signal (Non-Linear Phase):** When $\tau_g(\omega)$ varies with frequency, **phase dispersion** occurs. Harmonics shift asynchronously in time, resulting in severe waveform distortion, loss of symmetry, and peak amplitude attenuation.

6 Impact on Voltage Signal Characteristics

In power system harmonics, digital voltage buses, and analog instrumentation, non-constant group delay degrades key voltage parameters:

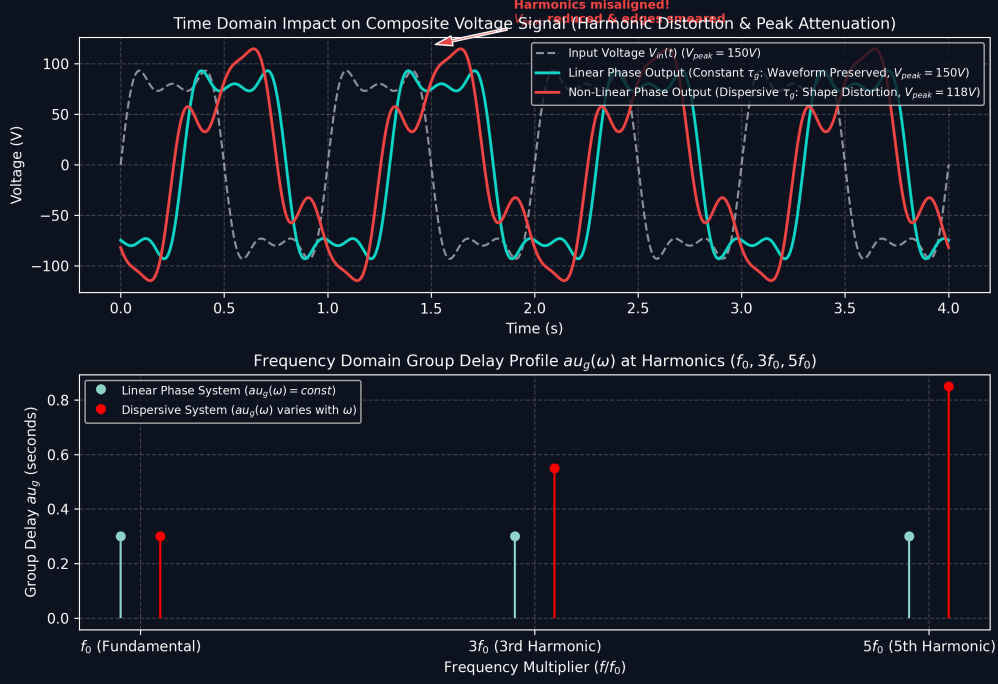


Figure 4: Impact of linear vs. non-linear group delay dispersion on a multi-harmonic composite voltage waveform.

1. **Peak Voltage Magnitude (V_{peak}) Degradation:** In multi-harmonic AC voltage signals $v(t) = \sum V_k \sin(k\omega_0 t + \phi_k)$, non-constant phase delays disrupt the constructive alignment of harmonic peaks, resulting in a reduced peak voltage amplitude or localized voltage overshoots.
2. **Edge Slew Rate ($\frac{dV}{dt}$) & Rise Time (t_r):** Sharp digital voltage steps suffer dispersion because high-frequency Fourier components are delayed differently from low frequencies. This smears the voltage step edge, reduces slew rate $\frac{dV}{dt}$, increases rise time t_r , and induces Inter-Symbol Interference (ISI) in data buses.
3. **Crest Factor ($CF = V_{peak}/V_{rms}$):** Even for an all-pass filter where total signal energy and V_{rms} are preserved, non-linear phase alters peak alignment, causing significant variations in the Crest Factor.
4. **Zero-Crossing Jitter & Synchronization Errors:** Non-uniform phase delay shifts zero-crossing timestamps of harmonic voltage components, introducing timing jitter in Zero-Crossing Detectors (ZCD), phase-locked loops (PLL), and SCR thyristor firing circuits.

7 Group Delay Derivation for First-Order Filter

Let $y[n] - ry[n-1] = x[n]$ with $|r| < 1$.

$$H(e^{j\omega}) = \frac{1}{1 - re^{-j\omega}} \quad (8)$$

$$\theta(\omega) = -\arctan\left(\frac{r \sin \omega}{1 - r \cos \omega}\right) \quad (9)$$

Let $u(\omega) = \frac{r \sin \omega}{1 - r \cos \omega}$. Using the derivative of $\arctan(u)$:

$$\frac{du}{d\omega} = \frac{r \cos \omega - r^2}{(1 - r \cos \omega)^2} \quad \text{and} \quad 1 + u^2 = \frac{1 + r^2 - 2r \cos \omega}{(1 - r \cos \omega)^2} \quad (10)$$

Combining these terms yields:

$$\frac{d\theta(\omega)}{d\omega} = -\frac{1}{1 + u^2} \frac{du}{d\omega} = -\frac{r \cos \omega - r^2}{1 + r^2 - 2r \cos \omega} \quad (11)$$

Thus:

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} = \frac{r \cos \omega - r^2}{1 + r^2 - 2r \cos \omega} \quad (12)$$

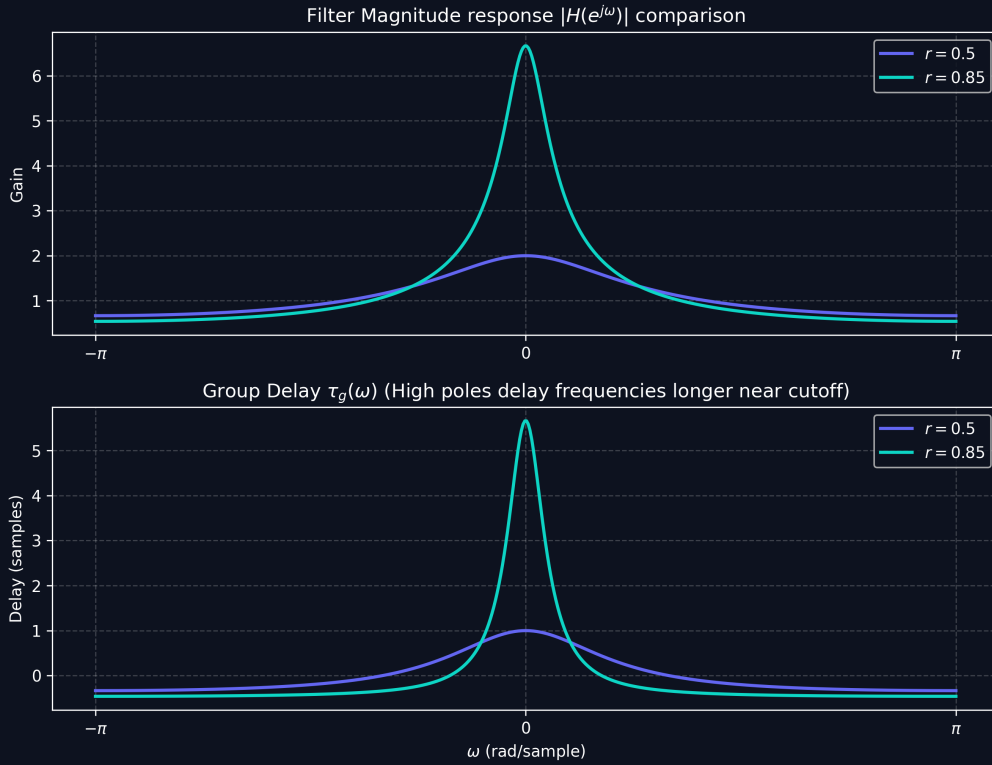


Figure 5: Filter magnitude and corresponding group delay curves for different pole choices.

8 Checkpoint & Numerical Review Questions

1. **Q1 (Pure Delay):** Find $\tau_p(\omega)$ and $\tau_g(\omega)$ for $h[n] = \delta[n - n_0]$.
Solution: $H(e^{j\omega}) = e^{-j\omega n_0} \implies \theta(\omega) = -\omega n_0$. Thus $\tau_p(\omega) = \tau_g(\omega) = n_0$ samples (constant delay $\rightarrow$ distortionless).
2. **Q2 (DC Group Delay):** For $r = 0.9$, evaluate $\tau_g(0)$.
Solution: $\tau_g(0) = \frac{0.9 - 0.81}{1 + 0.81 - 1.8} = \frac{0.09}{0.01} = 9$ samples.